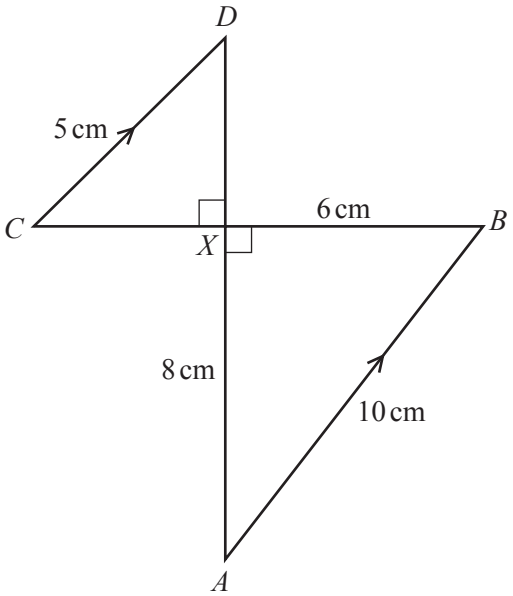
0580_w18_qp_43_Q8

8 (a)



NOT TO SCALE

In the diagram, AB and CD are parallel.
AD and BC intersect at right angles at the point X.
$AB = 10$ cm, $CD = 5$ cm, $AX = 8$ cm and $BX = 6$ cm.

(i) Use similar triangles to calculate DX.

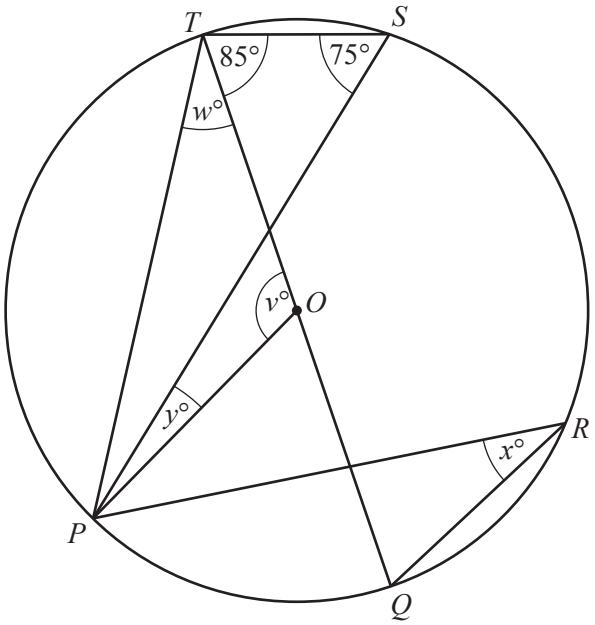
$DX =$ ..... cm [2]

(ii) Calculate angle XAB.

Angle $XAB =$ ..... [2]

0580_w18_qp_43_Q8

(b)



NOT TO SCALE

P, Q, R, S and T lie on the circle, centre O.
Angle $PST = 75°$ and angle $QTS = 85°$.

Find the values of v, w, x and y.

$v =$ .....

$w =$ .....

$x =$ .....

$y =$ ..... [6]

(c) Two containers are mathematically similar.
The surface area of the larger container is $226\,\text{cm}^2$ and the surface area of the smaller container is $94\,\text{cm}^2$.
The volume of the larger container is $680\,\text{cm}^3$.

Find the volume of the smaller container.

..... cm^3 [3]